

法一: $\frac{dx}{u_x} = \frac{dy}{u_y} = \frac{dz}{u_z}$

法二



$$\Rightarrow u_x dy = u_y dx$$

$$\Rightarrow -u_y dx + u_x dy = 0$$

$$\text{势函数} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

法四:
$$\begin{cases} u_x = \frac{\partial \psi}{\partial y} \\ u_y = -\frac{\partial \psi}{\partial x} \end{cases}$$

$$\psi = C \text{ 常数}$$

法三

法三: 法二

法三: 法二

$$\psi = C$$

法四: 法三

法四: 法三

$$\Rightarrow \frac{dx}{u_x} = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$

$$= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$